



WORCESTER POLYTECHNIC INSTITUTE

Monte Carlo and Resampling Methods Applied to Weak Gravitational Lensing Mass Reconstruction

MA 551 - Computational Statistics

May 4, 2026

Adam Field Department of Physics



SECTION 01

Motivation

Why lensing? Why statistics?

01

Weak Gravitational Lensing

Massive structures bend light from background galaxies, distorting their apparent shapes coherently across the sky.

The observable: galaxy ellipticity $e = e_1 + ie_2$

What we want: projected mass density $\kappa(\boldsymbol{\theta})$ (convergence)

- ▶ Direct probe of dark matter, dark energy equation of state
- ▶ Stage-IV surveys (Rubin, Euclid, Roman) are already underway
- ▶ Systematic errors at the **sub-percent level** will limit cosmological inference

The core problem: how do we propagate uncertainty, detect bias, and determine how much data we need?

The Statistical Problem

The mass reconstruction pipeline has two stages:

1. **Forward model:** $\kappa \rightarrow \gamma$ via a known linear FFT operator F
2. **Inverse:** recover $\hat{\kappa}$ from noisy shear observations γ_{obs}

Each source galaxy contributes an ellipticity measurement with intrinsic scatter $\sigma_e = 0.26$ per component (“shape noise”):

$$\gamma_{\text{obs}[i,j]} = (1 + m), \gamma_{\text{true}[i,j]} + \varepsilon[i,j], \quad \varepsilon[i,j] \sim \mathcal{N}(0, \sigma_e^2 / n_{\text{gal/pix}})$$

Key questions (all from MA 551):

- ▶ What is the reconstruction error, and how variable is it? (MC)
- ▶ Can we quantify uncertainty efficiently? (Antithetic, IS)
- ▶ What confidence intervals are valid? (Bootstrap, BCa)
- ▶ How many galaxies to detect 1% bias at 5σ ? (Power analysis)

SECTION 02

The Kaiser–Squires Algorithm

A validated R implementation

02

Kaiser–Squires Inversion

Shear and convergence are related in Fourier space via the KS kernel $D(\mathbf{k})$. Since $|D(\mathbf{k})| = 1$ for all $\mathbf{k} \neq 0$, the inverse is simply the conjugate. Optional Tikhonov regularization: replace $|D|^2$ with $|D|^2 + \lambda$.

$$\hat{\kappa}(\mathbf{k}) = \overline{D(\mathbf{k})}, \hat{\gamma}(\mathbf{k}), \quad D(\mathbf{k}) = \frac{k_1^2 - k_2^2 + 2ik_1k_2}{k_1^2 + k_2^2}$$

DC ambiguity: $D(0) = 0$ always, so the mean of κ is unrecoverable (mass-sheet degeneracy). The noiseless KS aperture mass (0.250) differs from the true value (0.346) by 9.6% due to this irreducible offset.

Analytical Validation: 10-Test Suite

The R implementation was validated against SMPy's KaiserSquiresMapper (production Python KS). Both compute the same algebraic expression.

Test	Result
FFT frequency convention matches <code>numpy.fft.fftfreq</code>	PASS
$ D(\mathbf{k}) = 1$ for all $\mathbf{k} \neq 0$ (machine precision)	PASS
Noiseless round-trip L_2 error $< 10^{-10}$	PASS
Linearity: $F(a\kappa_1 + b\kappa_2) = aF(\kappa_1) + bF(\kappa_2)$	PASS
Parseval / adjoint: $\langle F\kappa, \gamma \rangle = \langle \kappa, F^*\gamma \rangle$	PASS
B-mode purity $< 10^{-10}$ for physical κ	PASS
SMPy exact formula equivalence: max diff $< 10^{-14}$	PASS
Single-frequency (sinusoidal) analytical solution	PASS
DC non-contamination: adding constant to κ does not change γ	PASS
Peak amplitude recovered to 10^{-10} (DC-corrected)	PASS

SECTION 03

Simulation Study

Monte Carlo, antithetic variables, importance sampling

03

Simulation Setup

Grid: 32×32 pixels, 0.1 arcmin/pixel

Field: 3.2×3.2 arcmin²

True κ : Gaussian cluster, peak = 0.297,
 $\sigma_\ell = 0.5$ arcmin

Galaxies: $N_{\text{gal}} = 5,000$

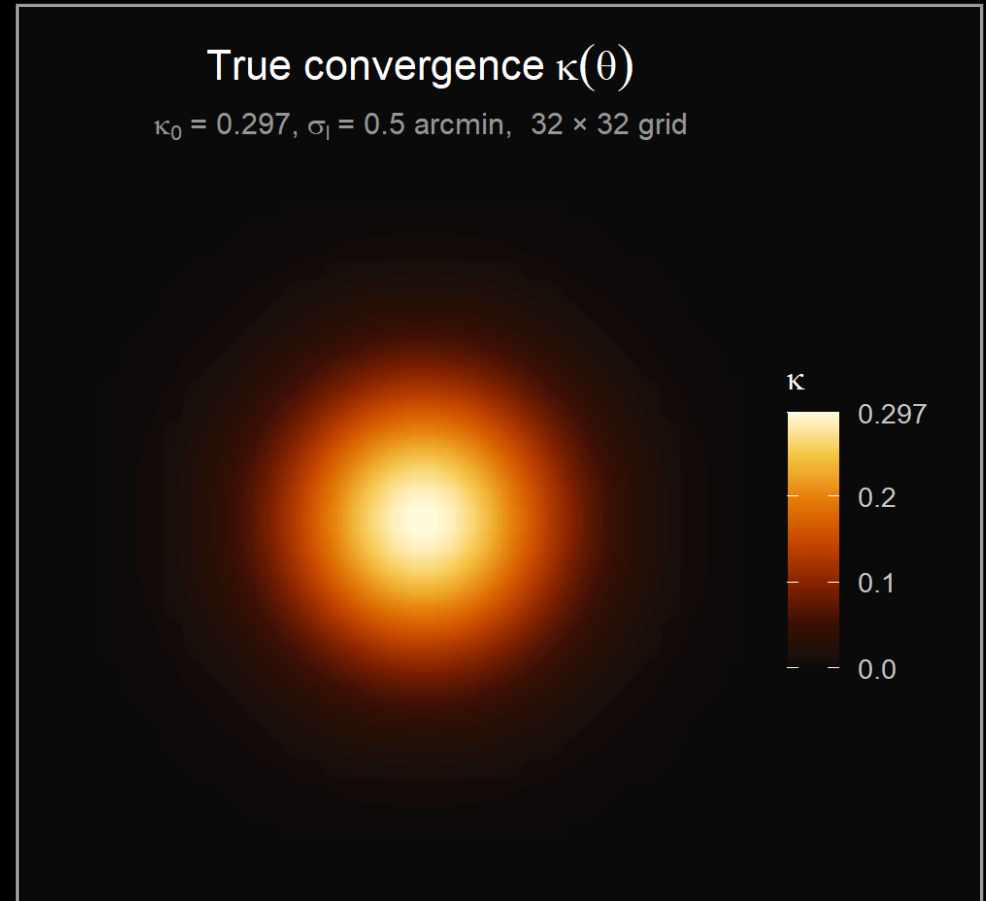
Noise: $\sigma_{\text{pix}} = 0.118$, $\max |\gamma| = 0.103$

SNR per pixel: ≈ 0.85 (noise-dominated)

Two scalar summaries:

- ▶ Peak $\hat{\kappa}$: maximum pixel value
- ▶ Aperture mass: $T_{\text{ap}} = \sum_{r \leq 0.8} \hat{\kappa} \delta\theta^2$

Aperture mass is preferred: it integrates over a region rather than taking the maximum of a noisy field.



Simple Monte Carlo: $B = 500$ Replicates

Estimate $\mathbb{E}[L_2(m=0)]$ and the distribution of scalar summaries under pure shape noise (Notes #12).

Summary	True value	MC mean	MC SD
L_2 error	θ ($m=0$)	1.521	0.045
Peak $\hat{\kappa}$	0.297	0.488	0.051
Aperture mass	0.250 [†]	0.251	0.015

[†] Noiseless KS value; true- κ aperture mass is 0.346 (DC offset 9.6%).

- ▶ **Peak kappa is severely upward-biased:** the maximum of 1024 noisy pixels is a noise spike, not the cluster. Bootstrap is not valid for this statistic.
- ▶ **Aperture mass is nearly unbiased** relative to the correct target (noiseless KS).
- ▶ L_2 error mean of 1.52 reflects the noise-dominated regime at $N_{\text{gal}} = 5000$.

Antithetic Variable Sampling (Notes #13)

Pair each noise draw ε with $-\varepsilon$. For monotone g :

$$\text{Var}(\hat{I}_{\text{anti}}) = \frac{\sigma^2}{2N}(1 + \text{Cor}(g(\varepsilon), g(-\varepsilon))) < \frac{\sigma^2}{2N}$$

Statistic	Variance reduction	Reason
Aperture mass	+100%	Linear in ε ; $\text{Cor} = -1$ exactly
Peak $\hat{\kappa}$	-4%	Maximum is not monotone
L_2 error	-100%	Degenerate (see below)

The degenerate case: $L_2(\varepsilon) = \|F^{-1}\varepsilon\| / \|\kappa_{\text{true}}\|$ depends only on $\|\varepsilon\|$, so $L_2(\varepsilon) = L_2(-\varepsilon)$ **exactly**. Antithetic pairs are identical; averaging them halves the effective sample size without any variance reduction.

This illustrates the key condition from Notes #13: **antithetic sampling requires the integrand to be monotone**. An even function in ε violates this.

Importance Sampling Over Bias (Notes #12, #13)

Estimate the bias-integrated error:

$$\mathbb{E}_{p[L_2(m)]} = \int L_2(m)p(m), dm, \quad p(m) = \mathcal{N}(0, 0.05^2)$$

Proposal: $q(m) = \text{Uniform}(-0.15, 0.15)$. Self-normalized weights $w_i = \frac{p(m_i)}{q(m_i)}$.

Method	Estimate (SE)
IS ($n = 50$, $B' = 100$)	1.518 (0.193)
Uniform grid	1.520

The IS and uniform estimates agree closely. This is itself informative: **the reconstruction error is nearly flat in m at this galaxy density.** The bias signal is buried in shape noise, so IS offers no efficiency gain. A more informative regime (higher N_{gal}) would show larger IS benefit near $m = 0$ where the prior concentrates.

SECTION 04

Bootstrap Inference

BCa confidence intervals and coverage

04

Parametric Bootstrap Setup (Notes #6, #7)

Parametric bootstrap: draw $B = 500$ fresh noise realizations from the known $\mathcal{N}(0, \sigma_{\text{pix}}^2)$ distribution. The noise model is known exactly.

BCa correction factors:

- ▶ **Bias correction** $\hat{z}_0 = \Phi^{-1}\left(B^{-1} \sum_b \mathbb{1}(\hat{T}_b^* < \hat{T})\right)$: adjusts for median shift
- ▶ **Acceleration** $\hat{a}$: estimated via **column jackknife** on the shear grid. Delete column j , recompute KS reconstruction. Requires $N = 32$ reconstructions (not N^2). For near-linear estimators, $\hat{a} \approx 0$.

Statistic	Obs.	Bias	SE	95% BCa CI
Peak $\hat{\kappa}$	0.251	+0.237	0.051	degenerate
Aperture mass	0.250	+0.000	0.015	[0.220, 0.278]

Peak BCa fails: every bootstrap replicate adds fresh noise, so the bootstrap peak **always exceeds** the noiseless reference. $\hat{z}_0 = \Phi^{-1}(0) = -\infty$.

Coverage Study: $n_{\text{outer}} = 200$ Replicates

For each outer replicate: generate one noisy observation, compute BCa and percentile CI, check coverage against the noiseless KS aperture mass target.

Interval type	Empirical coverage	Mean width
Percentile	97.5%	0.0586
BCa	97.5%	0.0585

Interpretation:

- ▶ Both achieve 97.5% at nominal 95%: slightly conservative, as expected when σ_{pix} is known exactly and the noise is exactly Gaussian.
- ▶ BCa $\approx$ percentile here because $\hat{a} = 0.006 \approx 0$: the aperture mass functional is nearly linear in the shear, so the estimator's SE is essentially constant and no skewness correction is needed.
- ▶ BCa would diverge from percentile for non-linear statistics (e.g., peak kappa) or at lower SNR where asymmetry becomes important.

SECTION 05

Power Analysis

How many galaxies to detect 1% bias at 5σ ?

05

Design: Why Unpaired?

Goal: detect multiplicative bias $|m| = 0.01$ via two-sample Welch t -test on B independent aperture mass replicates per condition.

The paired design is degenerate for linear statistics.

Since aperture mass is linear in the shear, the paired difference is:

$$D_b = T_{\text{ap}(m=0.01)} - T_{\text{ap}(m=0)} = 0.01 \times T_{\text{ap}}^{\text{KS}} = \text{const}$$

Every replicate gives exactly the same D_b , so $\text{sd}(D) = 0$ and $\text{NCP} = \infty$. This is the same linearity that caused the antithetic degeneracy.

Design: Why Unpaired?

Unpaired design: independent noise in each condition. Effect size is analytic:

$$\delta = |m| \times T_{\text{ap}}^{\text{KS}} = 0.01 \times 0.2504 = 0.00250$$

SD scales as $\frac{1}{\sqrt{N_{\text{gal}}}}$ from the shape noise model:

$$\sigma_{\text{ap}}(N_{\text{gal}}) = 0.0146 \times \sqrt{\frac{5000}{N_{\text{gal}}}}$$

Power Curve: Detecting $|m| = 0.01$ at 5σ

Non-centrality parameter and power for $B = 500$ replicates per condition, significance threshold $\alpha = 2.87 \times 10^{-7}$ (5-sigma).

N_{gal}	NCP	Power	σ_{ap}
1,000	1.21	< 0.001	0.033
2,000	1.71	0.001	0.023
5,000	2.70	0.008	0.015
10,000	3.83	0.096	0.010
20,000	5.41	0.610	0.007
50,000	8.55	> 0.999	0.005

Minimum N_{gal} for 80% power: $\approx 24,400$ (analytic uniroot).

MC verification at $N_{\text{gal}} = 5000$, $B = 500$: NCP = 2.30, power = 0.002 (analytic: 2.70, 0.008 – 15% gap from finite- B fluctuation, as expected).

SECTION 06

Conclusions

06

Summary of Findings

Method	Key finding
Simple MC	Aperture mass nearly unbiased; peak kappa severely biased by noise spikes
Antithetic	+100% reduction for aperture mass; degenerate for L_2 error (even function)
IS	No gain here: error flat in m at $N_{\text{gal}} = 5000$; bias buried in noise
Bootstrap	Aperture mass BCa well-behaved ($\hat{a} \approx 0$); peak kappa BCa undefined
Coverage	97.5% empirical at nominal 95%; BCa $\approx$ percentile for linear functionals
Power	24,400 galaxies for 80% power at 5σ detecting $ m = 0.01$, $B = 500$

Unifying theme: the linearity of the KS operator creates a consistent pattern. Linear statistics (aperture mass) are tractable for bootstrap and IS but degenerate for antithetic and paired designs. Non-linear statistics (peak) are biased and bootstrap-invalid. Statistical method choice is inseparable from the structure of the estimator.

Connections to MA 551 Curriculum

Every analysis in this project maps directly to course material:

- ▶ **Notes #3-4:** simulation-based hypothesis testing, critical values
- ▶ **Notes #6:** the bootstrap, empirical CDF, resampling
- ▶ **Notes #7:** BCa CI, jackknife, acceleration factor
- ▶ **Notes #12:** Monte Carlo integration, simple MC estimation
- ▶ **Notes #13:** antithetic variables (monotonicity condition), importance sampling (proposal design), control variates

The project is approximately 80% statistical analysis and 20% forward modeling. The astrophysics provides a concrete inverse problem where the statistical methods produce physically interpretable results.

Future work: GalSim-simulated galaxy images with realistic PSF convolution and metacalibration shear response correction, placing this analysis in a physically complete pipeline.

Acknowledgements

Prof. Fangfang Wang MA 551 course and feedback on the project proposal.

George Vassilakis Open-source Kaiser--Squires reference implementation.



WORCESTER POLYTECHNIC INSTITUTE

Thank you for your time!

Questions?